[CSL252]-Design and Analysis of Algorithms **2025-26-W**

TUTORIAL X

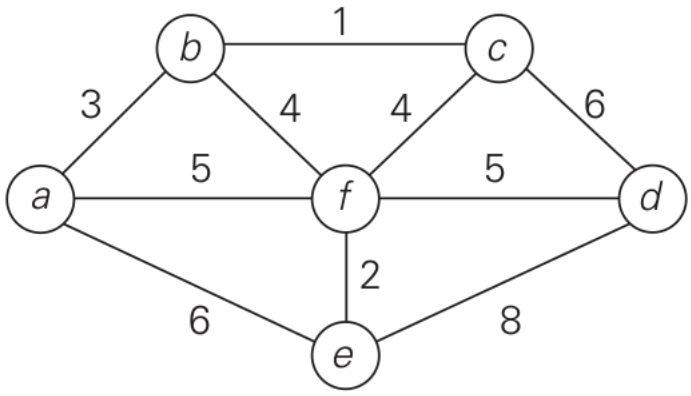
Date of the tutorial: **Mar 26, 2026**.

Due Date: **11:59 pm, Mar 31, 2026**.

Instructions. Same as Tutorial I and provide clear justification to True/False questions.

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1. Find the minimum spanning tree of the following graph using (a) Kruskal's Algorithm (b) Prim's Algorithm



2. TRUE/FALSE: Let G be an undirected graph with distinct edge weights, that is no two edges in the graph have equal weights.
   (a) The minimum weight edge is always be a part of a minimum spanning tree.
   (b) The maximum weight edge is never be a part of any minimum spanning tree.
   (c) The graph G has a unique minimum spanning tree.
3. TRUE/FALSE: Let G be an undirected edge-weighted graph.
   (a) The minimum weight edge is always be a part of a minimum spanning tree.
   (b) The maximum weight edge is never be a part of any minimum spanning tree.
   (c) The graph G has a unique minimum spanning tree.
4. TRUE/FALSE: Let e be a minimum-weight edge in a connected graph G, then e belongs to some minimum spanning tree of G.
5. Suppose we are given an undirected graph G with weighted edges and a MST T of G.

(a) Describe an algorithm to update the MST T when the weight of a single edge e is INCREASED.

(b) Describe an algorithm to update the MST T when the weight of a single edge e is DECREASED.

6. Describe and analyze an algorithm to compute the maximum-weight spanning tree of a given edge-weighted graph.

7. TRUE/FALSE: Justify your answer.

   (a) The path between any pair of vertices in a minimum spanning tree is necessarily a shortest path between those vertices in the original graph.

   (b) If all edges in the graph have distinct edge weights then the path between a pair of vertices in a minimum spanning tree necessarily a shortest path between the two vertices in the original graph.

8. Let G be an undirected, connected graph and all edge weights are distinct. A second best spanning tree T' is a spanning tree with smallest weight except for the MST.

   (a) TRUE/FALSE: the second-best minimum spanning tree is unique.

   (b) Give an algorithm to compute the second-best minimum spanning tree of G.